

ML 710 Project Instructions

Goal of Project: Apply course knowledge to parallelize an ML model

- You are encouraged to parallelize ML models from your own research
- If you don't use your own ML model, you must choose an ML model from a top-tier conference (NerulPS, ICML, CVPR, ACL, etc.)

Groups: Students are required to work in groups of 3

Compute: You need to use at least 2 GPUs. All students have access to HPC Labs SLURM, and you will learn how to use it during the course labs.

- If you have access to other GPUs from your advisor, you can use them instead
- You are strongly advised to **design your experiments so that each training run takes <1h**
- HPC Labs GPUs are shared among the whole student body, and you are expected to use them responsibly

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Deliverables: Your team will submit a PPT (or PDF slides) to the course Moodle for grading

- Maximum of 30 slides. Graders will not look at slides beyond 30.
- You are graded based on your individual contribution to the project. Each student must write their name on the slides that correspond to the strategies they implemented.
- Deadline is end of Week 14 (Sunday) at 23:59
- You do not need to submit a written or conference-style report, only the slides.

Grading: Your project grade will depend on (1) the sophistication and (2) number of different parallel/distributed strategies that you implement and evaluate. **Minimum requirements:**

- Each student must implement and evaluate at least 1 non-trivial parallel/distributed strategy
 - Groups with 3 students will therefore implement at least 3 non-trivial strategies
 - Model, pipeline, and hybrid-parallel strategies count as non-trivial strategies
 - Data-parallel strategies (excluding naïve ones such as PyTorch DDP) count as one-third of a non-trivial strategy. A student that chooses data-parallel strategies must implement at least 3 of them.
- Additionally, the team must include a naïve data-parallel strategy (e.g. PyTorch DDP) as a baseline to compare against. This baseline does not count towards the requirement of 1 non-trivial strategy per student.

Software: You may use any open-source tools to implement your parallel/distributed strategies.

- In your presentation, you must report which open-source tools you use